

MAT1002: Calculus II L01

Quiz 1 — 2025-2026, Term 2

Tuesday, 27 Jan 2026

1. (a) (5 points) Show that the sequence $\{a_n\}_{n=1}^{\infty}$ converges using the definition, where

$$a_n := \frac{n}{2n+1}.$$

Find its limit.

- (b) (5 points) Given that the sequence

$$\left\{ 2, 2 + \frac{1}{2}, 2 + \frac{1}{2 + \frac{1}{2}}, 2 + \frac{1}{2 + \frac{1}{2 + \frac{1}{2}}}, \dots \right\},$$

converges, find its limit.

Solution:

- (a) We compute

$$\left| a_n - \frac{1}{2} \right| = \left| \frac{n}{2n+1} - \frac{1}{2} \right| = \left| \frac{2n - (2n+1)}{2(2n+1)} \right| = \frac{1}{2(2n+1)}.$$

For all $n \geq 1$ we have $2(2n+1) = 4n+2 \geq 4n$, so

$$\left| a_n - \frac{1}{2} \right| \leq \frac{1}{4n}.$$

Given $\varepsilon > 0$, choose $N > \frac{1}{4\varepsilon}$. Then for all $n \geq N$,

$$\left| a_n - \frac{1}{2} \right| \leq \frac{1}{4n} \leq \frac{1}{4N} < \varepsilon.$$

Hence by the definition of limit $\lim_{n \rightarrow \infty} a_n = \frac{1}{2}$.

- (b) Let the terms be denoted $b_1 = 2$ and $b_{n+1} = 2 + \frac{1}{b_n}$ for $n \geq 1$. If the sequence converges to a limit L , then taking limits on both sides gives

$$L = 2 + \frac{1}{L}.$$

Multiply both sides by L :

$$L^2 = 2L + 1 \implies L^2 - 2L - 1 = 0.$$

Solving the quadratic,

$$L = \frac{2 \pm \sqrt{4+4}}{2} = 1 \pm \sqrt{2}.$$

Since every term of the sequence is positive, the limit must be nonnegative, so $L = 1 + \sqrt{2}$.

2. (a) (5 points) Discuss the convergence of the p -series $\sum_{n=1}^{\infty} \frac{1}{n^p}$. You may use without proof any standard results.
- (b) (5 points) For $p = 2$, using the integral test approximate the series with $N = 9$. Find an upper bound for the error. (Note: $s_9 = 1 + \frac{1}{4} + \cdots + \frac{1}{9^2} \approx 1.53977$.)

Solution:

- (a) Consider the function $f(x) = x^{-p}$ on $[1, \infty)$. For $p \leq 0$ we have $a_n = 1/n^p$ does not tend to 0, so the series diverges. For $p > 0$, f is positive and eventually decreasing; the integral

$$\int_1^{\infty} x^{-p} dx$$

converges exactly when $p > 1$ and diverges for $p \leq 1$. By the integral test the p -series converges if and only if $p > 1$, and diverges for $p \leq 1$.

- (b) Let $S = \sum_{n=1}^{\infty} \frac{1}{n^2}$ and denote $s_N = \sum_{n=1}^N \frac{1}{n^2}$ and the remainder $R_N = S - s_N$. Since $f(x) = 1/x^2$ is positive and decreasing on $[1, \infty)$, the integral test inequalities give

$$\int_{N+1}^{\infty} \frac{dx}{x^2} \leq R_N \leq \int_N^{\infty} \frac{dx}{x^2}.$$

Evaluate the integrals:

$$\int_t^{\infty} \frac{dx}{x^2} = \left[-\frac{1}{x} \right]_t^{\infty} = \frac{1}{t}.$$

Hence

$$\frac{1}{N+1} \leq R_N \leq \frac{1}{N}.$$

For $N = 9$ we have

$$\frac{1}{10} = 0.1 \leq R_9 \leq \frac{1}{9} \approx 0.111111,$$

so

$$1.53977 + 0.1 \leq S \leq 1.53977 + 0.111111,$$

i.e.

$$1.63977 \leq S \leq 1.650881.$$

Therefore the error after nine terms satisfies $R_9 \leq \frac{1}{9} \approx 0.111111$. (A tighter statement is $0.1 \leq R_9 \leq 0.111111$.)

3. Justify whether the following series converge absolutely, converge conditionally, or diverge.

(a) (3 points) $\sum_{n=1}^{\infty} \frac{2^n + 5}{3^n}.$

(b) (3 points) $\sum_{n=1}^{\infty} a_n$, where

$$a_n = \begin{cases} \frac{n}{2^n}, & n \text{ is odd,} \\ \frac{1}{2^n}, & n \text{ is even.} \end{cases}$$

(c) (4 points) $\sum_{n=1}^{\infty} (-1)^n \frac{n^2}{n^3 + 1}.$

Solution:

(a) Split the sum:

$$\sum_{n=1}^{\infty} \frac{2^n + 5}{3^n} = \sum_{n=1}^{\infty} \left(\frac{2}{3}\right)^n + 5 \sum_{n=1}^{\infty} \left(\frac{1}{3}\right)^n.$$

Both are geometric series with ratios $2/3$ and $1/3$ respectively (both < 1), hence they converge absolutely. In fact,

$$\sum_{n=1}^{\infty} \left(\frac{2}{3}\right)^n = \frac{2/3}{1 - 2/3} = 2, \quad 5 \sum_{n=1}^{\infty} \left(\frac{1}{3}\right)^n = 5 \cdot \frac{1/3}{1 - 1/3} = \frac{5}{2},$$

so the total sum equals $2 + \frac{5}{2} = \frac{9}{2}$.

(b) We show absolute convergence. Consider $|a_n|$. For even n we have $|a_n| = 1/2^n$, for odd n we have $|a_n| = n/2^n$. Use the root test (or compare with a convergent series):

$$\limsup_{n \rightarrow \infty} \sqrt[n]{|a_n|} = \frac{1}{2},$$

because for odd n , $\sqrt[n]{n}/2 \rightarrow 1/2$, and for even n the n th root is $1/2$. Since $1/2 < 1$, the series $\sum |a_n|$ converges, hence $\sum a_n$ converges absolutely.

(c) Let $u_n = \frac{n^2}{n^3 + 1}$. First note $\lim_{n \rightarrow \infty} u_n = 0$ and for sufficiently large n the sequence (u_n) is positive and decreasing (one can check by differentiation of $f(x) = x^2/(x^3 + 1)$ or compare $u_{n+1} < u_n$ for large n). Therefore by the alternating series test, the alternating series $\sum (-1)^n u_n$ converges. For absolute convergence, compare

$$u_n \sim \frac{n^2}{n^3} = \frac{1}{n} \quad \text{as } n \rightarrow \infty,$$

so $\sum u_n$ behaves like the harmonic series and diverges. Thus the alternating series converges conditionally (convergent but not absolutely convergent).